

# Bloom Filter-Based MPSI

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# Bibliography

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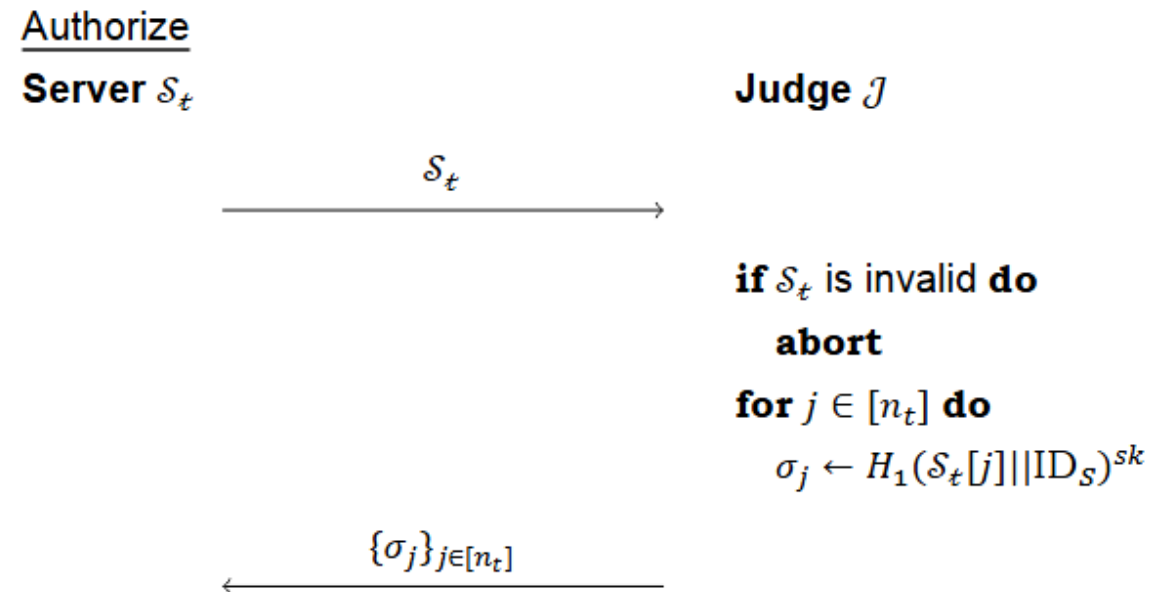
# Progress

- Start implementing APSI2+PAPSI2 / APSI3 / APSI4+PAPSI4 / **APSI5+PAPSI5**
- Writing!

# Progress

## APSI 5 – Lilika's Garbled Judge

- There can be collisions in  $H_1$  (which add to the FP probability), but they are negligible



# Progress

## APSI 5 – Lilika's Garbled Judge

- There can be collisions in  $H_1$  (which add to the FP probability), but they are negligible
- No encryption!

Intersect  
Server  $S_t$

Client 1  $S_1$

$s_1 \leftarrow \mathbb{Z}_q$

$S_1 \leftarrow g_2^{s_1}$

**for**  $j \in [n_1]$  **do**

$\hat{y}_{1,j} \leftarrow e(H_1(S_1[j] || \text{ID}_S), pk^{s_1})$

$\text{GBF}_1 \leftarrow \text{ComputeGBF}(\{\hat{y}_{1,j}\}_{j \in [n_1]})$

$S_1, \text{GBF}_1$

$S_i, \text{GBF}_i$

$S_{\text{agg}} \leftarrow \prod_{i=1}^{t-1} S_i$

**for**  $v \in [m]$  **do**

$\text{GBF}_{\text{agg}}[v] \leftarrow \prod_{i=1}^{t-1} \text{GBF}_i[v]$

$S_{\text{agg}}, \text{GBF}_{\text{agg}}$

$I = \emptyset$

**for**  $j \in [n_t]$  **do**

$\hat{x}_j \leftarrow e(\sigma_j, S_{\text{agg}})$

$\hat{y}_j \leftarrow \prod_{u=0}^k \text{GBF}_{\text{agg}}[h_u(\hat{x}_j)]$

**if**  $\hat{x}_j = \hat{y}_j$  **do**

$I = \{S_t[j]\} \cup I$

Output  $I$

Client  $i$   $S_i$

$s_i \leftarrow \mathbb{Z}_q$

$S_i \leftarrow g_2^{s_i}$

**for**  $j \in [n_i]$  **do**

$\hat{y}_{i,j} \leftarrow e(H_1(S_i[j] || \text{ID}_S), pk^{s_i})$

$\text{GBF}_i \leftarrow \text{ComputeGBF}(\{\hat{y}_{i,j}\}_{j \in [n_i]})$



# Next Week

- Continue implementing APSI2+PAPSI2 / APSI3 / APSI4+PAPSI4 / APSI5+PAPSI5
- Writing!

# Thank you!

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